

Things to Know

- Parseval's theorem has a different statement for CTFS, CTFT, DTFS, and DTFT. For CTFS, it is

$$\frac{1}{T} \int_0^T |x(t)|^2 dt = \sum_{k=-\infty}^{\infty} |X[k]|^2.$$

For CTFT, it is

$$\int_{-\infty}^{\infty} |x(t)|^2 dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} |X(j\omega)|^2 d\omega.$$

For DTFS, it is

$$\frac{1}{N} \sum_{n=0}^{N-1} |x[n]|^2 = \sum_{k=0}^{N-1} |X[k]|^2.$$

For DTFT, it is

$$\sum_{n=-\infty}^{\infty} |x[n]|^2 = \frac{1}{2\pi} \int_{-\pi}^{\pi} |X(e^{j\omega})|^2 d\omega.$$

- Convolving any signal with the step function is equivalent to integration. But to take care of the domain here's a simple rule.

For a signal $x(t) = f(t)u(t-a) + g(t)u(t-b) + h(t)u(t-c)$ with $a < b < c$, it's convolution with the step function is

$$x(t) * u(t) = \begin{cases} 0 & , t < a, \\ \int_a^t f(\tau) d\tau & , t > a, \\ \int_a^t f(\tau) d\tau + \int_b^t g(\tau) d\tau & , t > b. \\ \int_a^t f(\tau) d\tau + \int_b^t g(\tau) d\tau + \int_c^t h(\tau) d\tau & , t > c. \end{cases}$$

- If the gaussian integral $\int_{-\infty}^{\infty} e^{-at^2} dt = I$, then

$$\begin{aligned} I^2 &= \int_{-\infty}^{\infty} e^{-at^2} dt \cdot \int_{-\infty}^{\infty} e^{-a\tau^2} d\tau = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-a(t^2+\tau^2)} dt d\tau \\ &= \int_0^{2\pi} \int_0^{\infty} e^{-ar^2} r dr d\theta \\ &= 2\pi \int_0^{\infty} e^{-ar^2} r dr \\ &= 2\pi \cdot \frac{1}{2a} \\ &= \frac{\pi}{a}. \end{aligned}$$

Therefore, $I = \sqrt{\frac{\pi}{a}}$. Similarly, if the gaussian integral

$$\int_{-\infty}^{\infty} e^{-at^2} u(t) dt = \sqrt{\frac{\pi}{2a}}$$

- If the geometric series $\sum_{k=0}^n r^k = S$, then

$$\begin{aligned}
S &= 1 + r + r^2 + \cdots + r^n \\
rS &= r + r^2 + \cdots + r^n + r^{n+1} \\
S - rS &= 1 - r^{n+1} \implies S = \frac{1 - r^{n+1}}{1 - r}.
\end{aligned}$$

For a infinite geometric series $\sum_{k=0}^{\infty} r^k$

$$\sum_{k=0}^{\infty} r^k = \frac{1}{1 - r}.$$

As for the alternating geometric series $\sum_{k=0}^n (-1)^k r^k = S$

$$S = \frac{1}{1 + r}.$$

- The convolution $\sum_{k=0}^n \alpha^k \beta^{n-k}$ can be calculated as

$$\sum_{k=0}^n \alpha^k \beta^{n-k} = \beta^n \cdot \sum_{k=0}^n \left(\frac{\alpha}{\beta}\right)^k = \beta^n \cdot \frac{1 - \left(\frac{\alpha}{\beta}\right)^{n+1}}{1 - \frac{\alpha}{\beta}} = \frac{\beta^{n+1} - \alpha^{n+1}}{\beta - \alpha}.$$